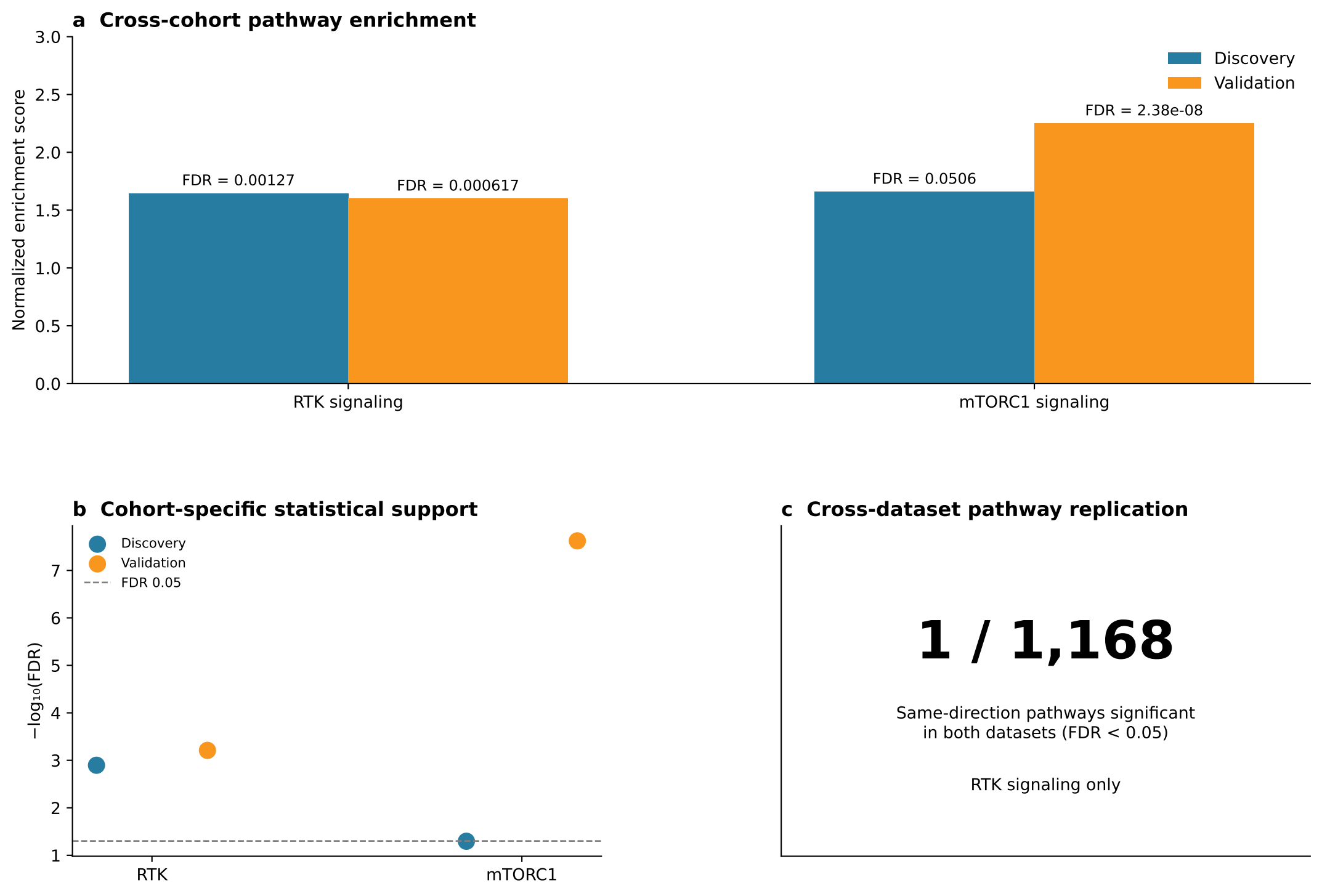
a Cross-cohort pathway enrichment
Normalized enrichment score
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Discovery
Validation
FDR = 0.00127
FDR = 0.000617
FDR = 0.0506
FDR = 2.38e-08
RTK signaling
mTORC1 signaling
b Cohort-specific statistical support
Discovery
Validation
FDR 0.05
−log₁₀(FDR)
7
6
5
4
3
2
1
RTK
mTORC1
c Cross-dataset pathway replication
1 / 1,168
Same-direction pathways significant
in both datasets (FDR < 0.05)
RTK signaling only